

Force and laws of motion: Source questions

IIT-JEE · 2007 · Paper I · Q3

Official past paper, faithful restatement of the original

Subject: Physics

Topic: Laws of Motion

Concept: Net force, not any one force, sets acceleration

Concept difficulty: Medium concept

Question difficulty: Multi-step reasoning

Family: Newton-law and observer-frame practice

Question type: Single-correct multiple choice

Source: IIT-JEE 2007 · Paper I · Q3

Provenance: Verified past-exam adaptation

Two equal masses m lie on a smooth horizontal surface and are joined by a light string of total length $2a$. The midpoint of the initially taut string is pulled vertically upward by a small constant force F . When the masses are separated by $2x$, determine the magnitude of the acceleration of either mass.

Conditions

- Horizontal surface is frictionless.
- String is light and remains taut.
- Each half-string has length a .

(A) $(F/2m) \cdot a / \sqrt{a^2 - x^2}$

(B) $(F/2m) \cdot x / \sqrt{a^2 - x^2}$

(C) $(F/2m) \cdot x / a$

(D) $(F/2m) \cdot \sqrt{a^2 - x^2} / x$

Hints from the source

1. How does the vertical displacement relate to half-string length and half-separation?
2. At the pulled midpoint, which components of the two equal tensions balance the applied force?
3. What horizontal force acts on either mass, and how does Newton's second law determine its acceleration?

Authored learning support

1. Sketch one half-string as a right triangle with horizontal projection x and vertical rise y .
2. At the pulled midpoint, which components of the two equal tensions balance F ?
3. After finding T , resolve only its horizontal component on either mass.

Answer and working

Option (B): $a_{mass} = F x / [2m \sqrt{a^2 - x^2}]$.

Authored reasoning route

1. **REPRESENT** Let the midpoint rise by y . String geometry gives $x^2 + y^2 = a^2$, so $y = \sqrt{a^2 - x^2}$.

Why valid: This follows from the stated physical or chemical model and the quantities preserved in the source-matched task.

Uses: stated source quantities

Establishes: intermediate result 1

2. **CONNECT** At the massless midpoint, vertical force balance gives $2T(y/a)=F$, hence $T=Fa/(2y)$.

Why valid: This follows from the stated physical or chemical model and the quantities preserved in the source-matched task.

Uses: stated source quantities

Establishes: intermediate result 2

3. **TRANSFORM** The horizontal force on either mass is $T(x/a)$, so its acceleration is $F x/[2m \sqrt{a^2 - x^2}]$.

Why valid: This follows from the stated physical or chemical model and the quantities preserved in the source-

matched task.

Uses: stated source quantities

Establishes: Option (B): $a_{\text{mass}} = F x / [2m \sqrt{a^2 - x^2}]$.

1. Let the midpoint rise by y . String geometry gives $x^2 + y^2 = a^2$, so $y = \sqrt{a^2 - x^2}$.
2. At the massless midpoint, vertical force balance gives $2T(y/a) = F$, hence $T = Fa/(2y)$.
3. The horizontal force on either mass is $T(x/a)$, so its acceleration is $F x / [2m \sqrt{a^2 - x^2}]$.

Independent check

As x approaches zero, the horizontal acceleration approaches zero because the string segments become nearly vertical.